

SQL Case Study

Restaurant Analytics — Jomato Dataset

Using SQL to analyze a food-delivery platform's restaurant data — covering user-defined functions, CASE logic, date functions, and ROLLUP.

- User-Defined Functions (STUFF)
- Aggregate Filtering with Subqueries (MAX)
- CASE Statements for Rating Classification
- Date & Math Functions — CEILING, FLOOR, ABS, GETDATE
- Grouping & Subtotals with ROLLUP

Step 1: Problem Statement & Tasks

Dataset: Jomato

About the dataset:

You work for a data analytics company, and your client is a food delivery platform similar to Jomato. They have provided you with a dataset containing information about various restaurants in a city. Your task is to analyze this dataset using SQL queries to extract valuable insights and generate reports for your client.

Tasks to be performed:

1. Create a user-defined functions to **stuff** the Chicken into 'Quick Bites'. Eg: 'Quick Chicken Bites'.
2. Use the function to display the restaurant name and cuisine type which has the maximum number of rating.
3. Create a Rating Status column to display the rating as 'Excellent' if it has more the 4 start rating, 'Good' if it has above 3.5 and below 4 star rating, 'Average' if it is above 3 and below 3.5 and 'Bad' if it is below 3 star rating and
4. Find the **Cell**, **floor** and **absolute** values of the rating column and display the **current date** and separately display the year, month_name and day.
5. Display the restaurant type and total average cost using **rollup**.

Step 2: Dataset Preview — Jomato.csv

Showing first 10 of 7,104 rows | 12 columns total (key columns shown below)

OrderId	RestaurantName	RestaurantType	Rating	No.of.Rating	AverageCost	CuisinesType	Area
1	L-81 Cafe	Quick Bites	3.9	48	400.0	Fast Food, Beverages	Byresandra,Tavarek
2	refuel	Cafe	3.7	37	400.0	Cafe, Beverages	Bannerghatta Road
3	Biryani Central	Casual Dining	2.7	135	550.0	Biryani, Mughlai, Chinese	Marathahalli
4	The Bbq	Casual Dining	2.8	40	700.0	BBQ, Continental, North Indi	Bellandur
5	The Bbq	Takeaway, Delivery	3.4	37	200.0	Mughlai, Biryani, Chinese, N	Whitefield
6	Italy	Casual Dining	4.1	305	700.0	Italian	Banashankari
7	North Parontha Hut	Takeaway, Delivery	2.8	40	300.0	North Indian	Indiranagar
8	1000 B.C	Quick Bites	3.2	49	300.0	Arabian, Sandwich, Rolls, Bu	Byresandra,Tavarek
9	1Q1	Casual Dining	3.7	41	450.0	Biryani, North Indian	Byresandra,Tavarek
10	11 to 11 Express Bir	Quick Bites	3.5	22	300.0	Biryani, Kebab	Electronic City

Step 3.1: Query 1: User-Defined Function — STUFF() to reformat Restaurant Type

```
USE JomatoDB;  
GO
```

```
SELECT TOP 10 *  
FROM dbo.Jomato;
```

--1. Create a user-defined function to change Quick Bites into Quick Chicken Bites

```
CREATE FUNCTION dbo.AddChicken (@RestaurantType VARCHAR(100))  
RETURNS VARCHAR(150)  
AS  
BEGIN  
    RETURN STUFF(@RestaurantType, 7, 0, 'Chicken ');  
END;  
SELECT dbo.AddChicken('Quick Bites') AS NewRestaurantType;
```

Step 3.2: Query 2: Using the Function — Restaurant with Maximum Ratings

--2. Use the function to display restaurant name and cuisine
--type having the maximum number of ratings

```
USE JomatoDB;  
GO
```

```
SELECT  
    RestaurantName,  
    CuisinesType,  
    RestaurantType,  
    No_of_Rating  
FROM dbo.Jomato;
```

```
USE JomatoDB;  
GO
```

```
SELECT  
    RestaurantName,  
    CuisinesType,  
    dbo.AddChicken(RestaurantType) AS ModifiedRestaurantType  
FROM dbo.Jomato  
WHERE No_of_Rating = (  
    SELECT MAX(No_of_Rating)  
    FROM dbo.Jomato  
)  
);
```

Step 3.3: Query 3 & 4: CASE Statement for Rating Status + Date/Math Functions

--3. Rating Status

```
SELECT
    RestaurantName,
    Rating,
    CASE
        WHEN Rating > 4 THEN 'Excellent'
        WHEN Rating > 3.5 THEN 'Good'
        WHEN Rating > 3 THEN 'Average'
        ELSE 'Bad'
    END AS [Rating Status]
FROM dbo.Jomato;
```

--4. Find the Ceil, floor and absolute values of the rating column and

--display the current date

--and separately display the year, month_name and day.

USE JomatoDB;

GO

```
SELECT
    Rating,
    CEILING(Rating) AS Ceiling_Rating,
    FLOOR(Rating) AS Floor_Rating,
    ABS(Rating) AS Absolute_Rating,
    GETDATE() AS TodayDate,
    YEAR(GETDATE()) AS CurrentYear,
    DATENAME(MONTH, GETDATE()) AS Month_Name,
    DAY(GETDATE()) AS Current_Day
FROM dbo.Jomato;
```

Step 3.4: Query 5: Restaurant Type & Total Average Cost using ROLLUP

--5.Display the restaurant type and total average cost using rollup.

USE JomatoDB;

GO

SELECT

ISNULL(RestaurantType, 'Grand Total') AS RestaurantType,|

AVG(AverageCost) AS Total_Average_Cost

FROM dbo.Jomato

GROUP BY ROLLUP(RestaurantType);